

JMO – 1999

Max Mark – 100

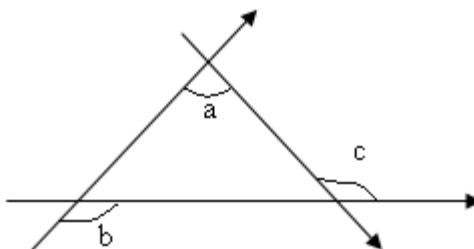
Time – 3 Hrs

Instructions:

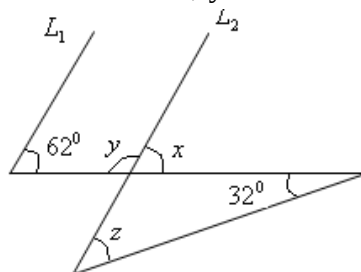
- Calculators (in any form) and protractors are not allowed.
- Ruler and compass allowed.
- Answer all questions.

1.

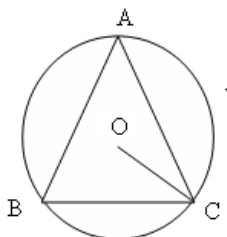
- Write in ascending order: $2^{32}, 4^{15}, 8^{11}, 16^8, 32^6$.
- A's monthly income is 20% more than B's monthly income. Find by how much % B's monthly income is less than that of A?
- Find the largest prime factor of 10626.
- The sum of four consecutive numbers is 90. Find the numbers.
- Today i.e. 9th January 2000 is a Sunday. What day will be on 27th February this year?
- If $x*y = x+y+1$, Evaluate $20*(-5)$.
- In the given diagram $m\angle a = 35^\circ$. Find the value of $m\angle b + m\angle c$.



- In the given diagram $L_1 \parallel L_2$. Find the value of x, y and z.



- In the given diagram $m\angle A = 60^\circ$, find the value of $m\angle OCB$.

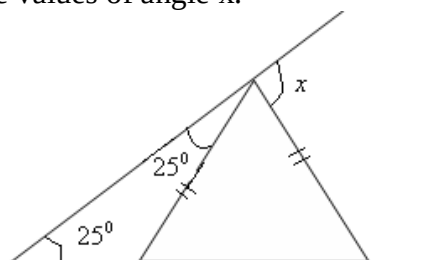


- 4% of 400 – 2% of 800 =?

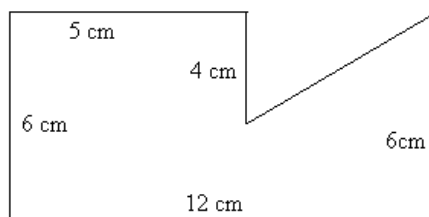
2.

- At what rate per annum, in simple interest will a sum of money double itself in 12 years?
- The ration between the ages of Gayatri and Savitri is 6:5 and the sum of their ages is 44 years. What will be the ratio of their ages after 8 years?

- (c) Fill up the missing digits by * in 812^{**} , so that the number will be a perfect square.
 (d) From the given figure find the values of angle x .



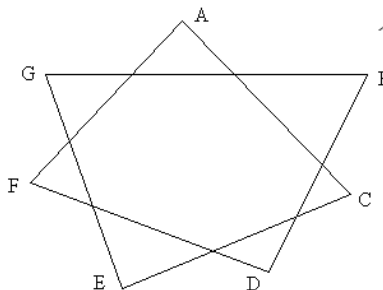
- (e) The areas of a square in cm^2 is same as its perimeter in cm. Find the length of the sides of the square in cm.
 (f) A owes Rs. 125 to B, B owes Rs 90 to C, C owes Rs 70 to D and D owes Rs. 67 to A. How can they settle the account where only A has to pay B, C and D?
 (g) A man purchased 10 bananas but demand 5 extra bananas with the same money. The shopkeeper told him "I would incur a loss of 20% by doing that". Find out the % of profit made by the shopkeeper?
 (h) Which will be more profitable to buy an article (i) with successive commission of 25% and 15% or (ii) with successive commission of 30% and 10%?
 (i) Without doing actual multiplication find the product of 358796 and 999.
 (j) Rs 300 is divided in the ratio 2:3. The larger part then divided into two parts in the ration 3:2. Fins out the smallest part.
- 3.
- (a) The difference in the compound and simple interest of certain sum in 2 years at the rate of 5% per annum is $1\frac{1}{4}$ rupees. Find the sum.
 (b) At 6 O' clock in the morning a train leaves Bhubaneswar for Rourkela while another train leaves Rourkela for Bhubaneswar at the same time. It takes one train 8 hours and the other train 22 hours to make the trip. Both maintain constant speed. At what time do they cross each other?
4. Three persons started playing a game and they decided that at the end of each game the loser would pay to each of the two winners an equal amount of money that each winner had with him at that time. They played three games and each one of them had lost once. At the end of these games each was left with Rs 24. Find the amount of money each had at the start of the play.
5. If abc is three digit number such that $abc = (bc)^2$, $bc = c^2$, where bc is a two digit number. Find out the values of a , b and c .
- 6.
- (a) A person gave his wife $\frac{1}{3}$ rd portion of his property and he distributed the rest of his property equally among all his sons. It was found that his wife got three times more than each of his sons. How many sons the man has?
 (b) In the given figure lengths of sides are marked. What is the area of the figure?



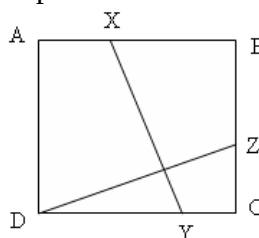
7. Two pipes A and B can separately fill up a water tank in $7\frac{1}{2}$ and 5 minutes separately and a third pipe C can carry off 14 litres of water per minute. If all the pipes are opened simultaneously when the tank is full of water it is emptied in one hour. How many litre of water does the tank hold?

8.

- (a) In the figure below, find $m\angle A + m\angle B + m\angle C + m\angle D + m\angle E + m\angle F + m\angle G = ?$.



- (b) If $XY \perp DZ$ in the square ABCD, then prove that $XY = DZ$.



9.

- (a) In continuing n colored balls some red and some black, it was found that the first 49 out of 50 counted were red. There after 7 out of every 8 counted were red. If in all 90% balls counted were red, then what will be the value of n?
- (b) There were 35 boys and girls. Every boy was given Rs 5 and every girl was given Rs 3. If the total amount given to them was Rs 145, determine the number of boys and girls.

10.

- (a) Find the largest number such that 500, 674 and 790 will leave the same remainder when divided by this number.
- (b) How many digits in total have been used to number the pages of a book from 1 to 400?

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